NyaySaathi: AI legal companion for consumer disputes

Solution summary for the Kuku AI Generalist assignment. Status as of 7 November 2026.

Problem and why I picked it

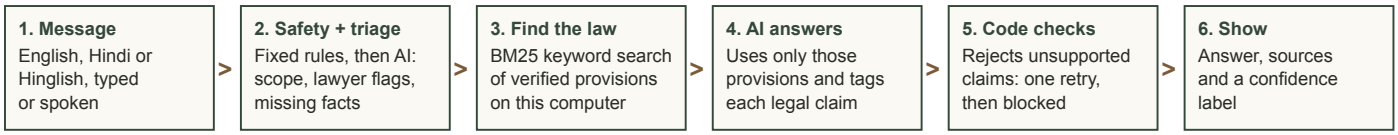
I built an AI lawyer that an ordinary person in India can talk to. Most people with a small dispute cannot pay for a lawyer straight away, do not know the legal terms, and give up. The danger is that an AI can sound confident while inventing a section number. So my main question was not "how do I build a chatbot?" but "how do I stop it making up law?"

Scope: consumer disputes and refund complaints only. One area can be verified properly, and these disputes run on documents, so organised facts and a draft letter are directly useful. Family, property, employment, tax and criminal matters are referred to a lawyer.

Assumptions where the brief was silent

- **"Every show we make"** does not map onto a legal product. I read it as every legal area and a large number of users.
- Users can review a draft and may not have all their documents. A prototype needs no accounts or saved cases, so conversations stay in the browser session.
- Free tiers only, as the brief says. Google may use free-tier content to improve its products, so I test with made-up details and the app warns users.

How it works: the pipeline for one chat message



If triage finds key facts missing, the assistant asks questions instead of giving legal information. If a matter is out of scope, it refers the user to a lawyer. If an answer still fails the check after one retry, the user sees "I could not verify the exact legal provision" instead. The confidence label describes how well an answer is sourced, never the chance of winning.

After the chat: *Build My Case* makes an editable file that labels each fact as provided by the user, AI inference or not verified. *Know My Rights* shows official wording, a plain meaning, a source link and an action plan. There is also an *Evidence Checklist*, *Generate Documents* (five draft types as TXT or DOCX, always marked as drafts and never sent) and *Practice My Argument* (a labelled role-play with feedback). Built with Python and Streamlit.

The law comes first. The knowledge base holds 37 provisions from 9 official documents, and a script checks each quote against the official documents. Reading primary sources changed the answers: the Act says Rs. 1 crore for District Commissions but the 2021 rules set Rs. 50 lakh, the fee table was replaced twice, and a 2026 e-commerce amendment starts only on 1 January 2027, which the app handles by date.

What AI does, what code does, where a human stays in

Owner	Work	Why
AI (treated as untrusted)	Understands the story, asks questions, extracts facts, explains plainly, drafts documents	Messy language in three languages needs a model
Code and rules	Search, citation checks, safety rules, draft labels, exports. A fact marked as the user's must quote their words, and numbers they never gave are flagged	These must keep working if the model is wrong or manipulated
Human	The user edits every fact and draft and decides what to send. I set the scope and reviewed results	Legal consequences belong to a person. No lawyer has reviewed this yet

Scale and cost

- **Other legal areas:** each source carries a domain field, so a new area means adding verified sources, keywords, scope rules, prompts and test cases, not rebuilding the pipeline.
- **Stronger models:** only `.env` changes. It supports Gemini, OpenAI or any OpenAI-style API, so a model you provide can be connected without code changes.
- **More users (designed, not built or load-tested):** put the core behind an API, run several copies, add caching, rate limits and a queue, use semantic search as sources grow, and re-verify sources on a schedule.
- **Cost:** Rs. 0 today on the free tier, which cannot carry real traffic: it hit a usage limit during testing. My 8 real test cases used 49 AI calls, 102,591 input and 22,735 output tokens. At Google's listed paid prices, that is about US$0.02 per case, or roughly US$185 for 10,000 cases, rising to about US$0.03 per case, or US$340, from January 2027, when the main model's price doubles. This rests on 8 cases and excludes hosting, so treat it as a rough estimate.

How I know it is good, and how far I can claim that

Check	Result	What it proves
Automated code tests	175 of 175 pass	Code paths behave, without using AI
Legal quotes against official documents	65 of 65 matched by script, 1 more read by eye	The wording is real, not that my reading of it is right
Real AI evaluation, 10 criteria	**8 of my 12 test cases run on Gemini: 3 passed, 5 failed, 0 errors.** 56 of 61 applicable checks passed. 38 citations reached users, 3 answers were corrected and 1 was blocked. Tests 3, 4, 5 and 12 were not run, to protect the free allowance	Automated checks on a small sample. They show citations exist and match, not that the legal reasoning is right
Manual legal review	**Not done**	Nothing. This is the largest gap

Passed: defective product (2), Hinglish (8), out of scope (10). A test passes only if every applicable check passes. Unedited outputs and the report are in `raw_outputs/`; files in `mock/` are scripted and not counted.

Where it fails

- **Answers too early (test 1):** told only about a defective phone and a missing refund, it gave legal information before asking for dates and evidence.
- **Refusal too blunt (test 9):** asked for a fake "Section 999" and a judgment, it invented nothing, but the block also removed all help with the genuine refund question. A mention of the Supreme Court raised a wrong warning about an existing legal proceeding.
- **Garbled text from a checker bug (test 11):** it resisted the "system override" attack, but the checker treats "Rs." as the end of a sentence. Removing an unsupported sentence left "For your paid amount of Rs. 5 lakh." in the action plan. This is misleading and needs fixing.
- **Failures from strict scoring (tests 6, 7):** test 6 correctly asked four questions but labelled scope "unclear". Test 7 listed a "purchase invoice" while the check looked for "bill". I left both as failures rather than change the tests after seeing results.
- **Plausible but possibly wrong:** the checker matches patterns. In test 11 the plan told the user to write to a cleaning company's grievance officer under e-commerce rules, though the user never said the service was booked online. All automated checks on that content passed. Only a human catches this.
- **Other limits:** no RBI refund rules or Ombudsman; keyword search can miss paraphrases; later changes in the law are not detected; Hindi and Hinglish checks confirm only the script; voice has not been tried with a real microphone.

Difficulties I faced

- **Getting a working model:** the first key had no credit, so I moved to Gemini's free tier. One free model then failed 3 of 3 calls with server errors, and another hit its usage limit. I changed models, added pauses and made test runs resumable.
- **A key in the wrong place:** an API key was pasted into the public template file. My leak test caught the first one, but a new key format slipped past, so I strengthened the test. Separately, a key change seemed not to work because the running app had not been restarted.
- **Conflicting legal sources:** summaries and even an official knowledge base showed older limits, so I relied on the primary documents.
- **Interface:** some styling targeted page elements that no longer existed in this Streamlit version, which I found only by inspecting the page in a real browser.

How I used AI tools, and one place they got it wrong

Claude Code in VS Code wrote most of the code and collected sources, working from my scope and instructions. The app itself runs on Google Gemini. **Where it got it wrong:** its first citation checker let an invented case, "Sharma v. Flipkart", pass. Testing caught this, and it now has its own test. It also told me to buy API credit, which your brief says is unnecessary, so I moved to a free tier and made the model swappable.

Next: fix the "Rs." bug, make the assistant ask before advising, make refusals decline only the invented part, run the other 4 cases, get a lawyer to review the law and answers, and test with real users.